



Semester 2 Examinations 2013/2014

Exam Code(s) 1MF, 1SD, 1MDM
Exam(s) Masters in Software Design & Development
Higher Diploma in Applied Science (Software Design & Development)
Masters in Digital Media

Module Code(s) CT870
Module(s) Internet Programming

Paper No. I
Repeat Paper

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Instructions: Answer any three questions.
All questions carry equal marks.

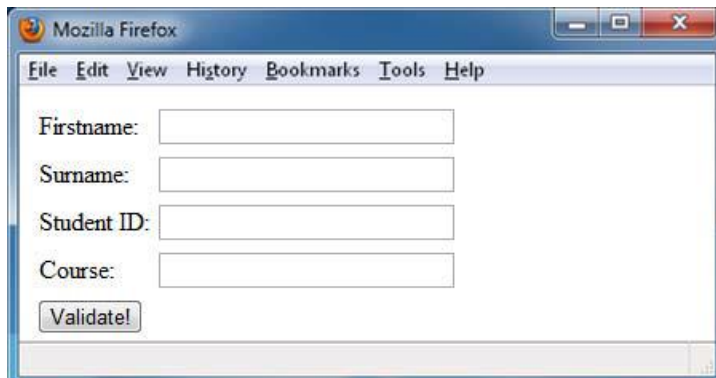
Duration 2 hours
No. of Pages 5
Department(s) Information Technology
Course Co-ordinator(s)

Requirements:

MCQ Release to Library: Yes ☐ No ☐
Handout
Statistical/ Log Tables
Cambridge Tables
Graph Paper
Log Graph Paper
Other Materials

Q1.

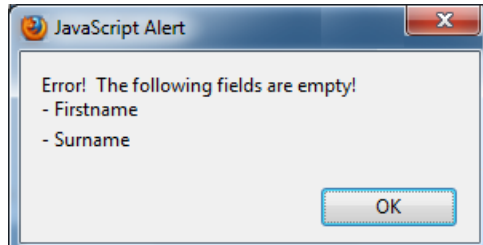
- a) Compare and contrast **Cookies** and **Session Variables** as methods of data storage. Which approach is more efficient? Which is more secure? Which is better for storing data for long periods? Use PHP examples to illustrate how information can be stored and retrieved using each. [8 marks]
- b) Write HTML code to produce the form as shown below.



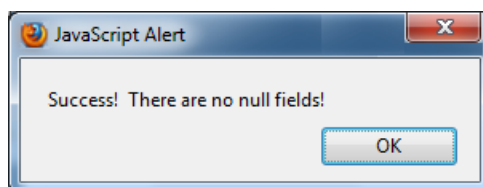
The screenshot shows a Mozilla Firefox browser window. The address bar is empty. The menu bar includes File, Edit, View, History, Bookmarks, Tools, and Help. The form contains four text input fields labeled 'Firstname:', 'Surname:', 'Student ID:', and 'Course:'. Below the 'Course:' field is a button labeled 'Validate!'.

[3 marks]

When the Validate button is clicked, use client side validation to ensure that no fields are left null. If any fields are left null, an alert box similar to below should appear, informing the user which fields were left empty. (NB: It is not necessary to validate for data type, merely for null (blank) strings).



If all fields are completed, an alert box similar to the one shown below should appear



[5 marks]

- c) Explain the use of the Javascript function **window.setTimeout()** for controlling animations in a webpage. Use sample code to illustrate your explanation.

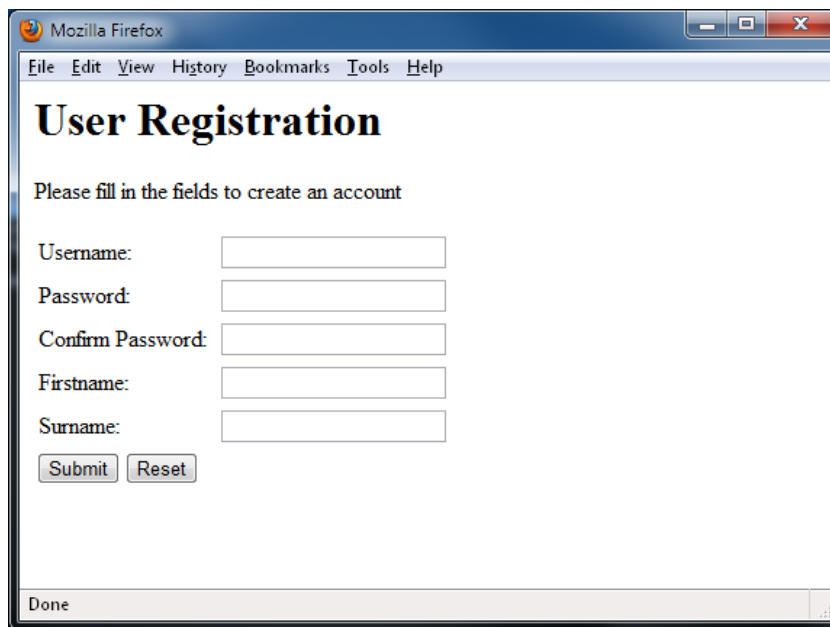
[4 marks]

Q2.

Create an HTML user registration form. The form will accept a user's registration details, and be posted to a PHP page which will insert the data into a MySQL database. Assume a MySQL database exists on the webserver, with a table called 'users' containing the following fields:

Fieldname	Datatype	Description
id	int(10)	A unique key for each user
username	varchar(50)	The user's username
password	varchar(50)	The user's plaintext password
firstname	varchar(50)	The user's first name
surname	varchar(50)	The user's surname

Write HTML code to produce a form similar to the one below [4 marks]



Ensure that the registration form is validated server side so that the following conditions are met:

- No fields are null
- The desired Username does not already appear in the database
- The Password and Confirm Password fields contain the same text [7 marks]

In the event of errors, messages should be displayed to the user highlighting the problematic field(s). Valid entries should be automatically filled into the text boxes so that they do not have to re-enter any fields that passed validation. [5 marks]

On successful validation of the form, the PHP script should insert the new details into the users table in the database. The user should also be presented with a message to confirm that the registration has completed successfully [4 marks]

Q3.

- a) Write code for a CSS External Stylesheet (.css file) which defines the following styles:
- All <H1> elements should use 16 point sized text, coloured green
 - All elements whose class is 'blue' should use blue text
 - All other elements should use red text
 - All elements should use the Times New Roman font
 - All <p> elements should align their text centrally
 - All <div> elements have borders where the top border = 10 pixels, the bottom border = 5 pixels, the left border = 20 pixels, and the right border = 1 pixel in width

[6 marks]

- b) An online bookshop uses a MySQL database which contains a table called 'books' with the following schema:

<u>Fieldname</u>	<u>Datatype</u>	<u>Description</u>
id	int(10)	A unique key for each book
title	varchar(50)	The title of the book
price	float	The price of the book (in Euro)
quantity	int(10)	The stock level, i.e. quantity in the shop
image	varchar(50)	The filename of an image of the book's artwork
author_id	int(10)	A foreign key to the authors table

The database also contains a table called 'authors', defined as follows:

<u>Fieldname</u>	<u>Datatype</u>	<u>Description</u>
id	int(10)	A unique key for each author
name	varchar(50)	The author's name
country	varchar(50)	The country that the author lives in

- Write PHP code to connect to the database and select all books written by the author whose id is 55
[3 marks]
- Extend this code so that the resulting dataset is displayed in an HTML page, with the details neatly laid out using a table. You can assume that the artwork images exist on the server, in a subfolder called 'images'
[5 marks]
- Write PHP code which receives a country (string) by query, (either by a POSTed form or a querystring). The PHP should display all book titles and author names where the author lives in the specified country. You do not need to construct the HTML page which posts this data, and you do not need to display any book details other than title.

[6 marks]

Q4.

- a) How can Javascript and DHTML be used to perform a simple animation which moves an image across the screen? Illustrate your answer with sample code. [4 marks]

- b) Consider the “7 Times Table” pictured below.

Times Tables!

1 x 7 =	7
2 x 7 =	14
3 x 7 =	21
4 x 7 =	28
5 x 7 =	35
6 x 7 =	42
7 x 7 =	49
8 x 7 =	56
9 x 7 =	63
10 x 7 =	70

- Write the HTML code necessary to display this table on a webpage, paying attention to styling of the table and text [3 marks]

- Explain in general terms, how might Javascript be used to automatically produce such a “times table” on a webpage when the user has typed a number into the text-box? What role might a <DIV> tag have in this process? [2 marks]

- Write suitable Javascript or PHP code to achieve this automatic “times table” production, noting any assumption made about the HTML form which contains the text box. [5 marks]

- c) Assuming that you have been supplied the three .gif images shown below:

- Write suitable HTML/CSS code to display the “one.gif” image in the centre of the screen. [2 marks]
- Write suitable Javascript code to achieve the following: (i) when the image is clicked it should be replaced with “two.gif”. (ii) When the image is clicked again it should be replaced with “three.gif”. (iii) On the third click, “one.gif” should be used again. Note that all changes should be done on the client using JavaScript, i.e. a new HTML page should not be requested from the server with each click. [4 marks]

